

Suriya Narayanan Lakshmanan

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EDUCATION

CMU, ROBOTICS INSTITUTE

MASTER'S IN COMPUTER VISION

Advised by Prof. Srinivasa Narasimhan

Dec 2018 | Pittsburgh, PA

Cum. GPA: 3.83/4.0

NIT, TIRUCHIRAPPALLI

BACHELOR OF TECHNOLOGY IN ELECTRICAL AND ELECTRONICS ENGINEERING

May 2014 | Tiruchirappalli, India

Cum. GPA: 8.8 / 10.0

COURSEWORK

Intro to Machine Learning

Intro to Computer Vision

Math fundamentals for Robotics

Visual Learning and Recognition

Geometry based maths in Vision

Computational Photography

Algorithms and Data Structures

Operating Systems

Object Oriented Programming

Digital Signal Processing

SKILLS

PROGRAMMING

C • C++ • CUDA • Python • Matlab

• OpenCL • $\text{\LaTeX}$

LIBRARIES

OpenCV • PyTorch (Python and C++)

• ROS • PCL • Numpy • TensorFlow

(Python and C++) • scikit-learn

OPERATING SYSTEMS

Linux • Windows • Android

OTHER SOFTWARE

Git • Microsoft Office • GIMP

EXPERIENCE

NVIDIA CORP. | SENIOR COMPUTER VISION ENGINEER

Oct 2022 - Present | Santa Clara, CA, USA

- Accelerated multiple AV modules using CUDA anywhere between 2x and 5x
- Produced and deployed DNN models for lidar based obstacle detection
- Working on DNN models for hazards detection through sensor fusion

NVIDIA CORP. | COMPUTER VISION ENGINEER

Feb 2022 - Sep 2022 | Bangalore, India

- Accelerated hazards detection code using CUDA achieving 18x speedup and can run under 3ms per LIDAR frame

NVIDIA CORP. | COMPUTER VISION ENGINEER

Jan 2021 - Jan 2022 | Santa Clara, CA, USA

- Prototyped, designed and implemented hazards detection algorithm for small obstacles using LIDAR

CYNGN INC (SELF-DRIVING VEHICLE COMPANY) | PERCEPTION ENGINEER

Feb 2019 - Oct 2020 | Menlo Park, CA, USA

- Designed and implemented traffic light detection system from scratch consisting of components such as traffic light detection, traffic light recognition and tracking
- Built 2d object detection system using PyTorch with BIFPN, Swish activation and deployed using TorchScript
- Evaluation of lidar based SLAM algorithms for indoor applications
- Analyzed the depth error of stereo and came up with an optimal baseline width taking into account target working range, camera FOV and permissible blindspot
- Designed and implemented software based Lidar-Camera synchronization system that is capable of operating at 10Hz under certain configurations

SAMSUNG RESEARCH AMERICA | COMPUTER VISION RESEARCH INTERN

Think Tank Team

May 2018 - August 2018 | Mountain View, CA, USA

- Developed human pose datasets for proprietary imaging sensors using unsupervised domain adaptation
- Developed a deep learning based human pose estimation network that can predict human poses on frames obtained from the proprietary imaging sensor
- Deployed the above developed network by creating a C++ application using TensorFlow APIs for building and executing the deep learning graph

TEXAS INSTRUMENTS | SOFTWARE ENGINEER

July 2014 - June 2017 | Bangalore, India

- Improved accuracy of TI CNN model for driver drowsiness detection by **2x**
- Improved Adaboost classifier for object detection yielding **10%** more true detections. [*Efficient object detection and classification on low power embedded systems, ICCE 2017*]
- Developed a set of Image Processing modules. [*Understanding the Performance Benefit of Asynchronous Data Transfers, HiPC 2015*]
- Accelerated a set of OpenCV functions using OpenCL and DSP which was released as part of TI Vision SDK. [*Understanding the Performance Benefit of Asynchronous Data Transfers in OpenCL Programs Executing on Media Processors, HiPC 2015*]
- Accelerated OpenCV using OpenCL, boosting performance by **3x over ARM A15**
- Released the above accelerated functions as applications in TI Vision SDK

TEXAS INSTRUMENTS | COMPUTER VISION INTERN

May 2013 - July 2013 | Bangalore, India

- Improved an existing homography based Ground Plane Detection by **10%**. [*Ground plane detection, Patent 2017*]. [*Improved ground plane detection in real time systems using homography, ICCE 2014*]

ACADEMIC PROJECTS

UNSUPERVISED SEGMENTATION DATA GENERATION

October 2018 - December 2018 | CMU, Pittsburgh

Interpolated semantic segmentation labels to frames in between key frames in NYU depth dataset v2 videos using dense depth as supervision for intermediate frames improving mIoU by 2% over FCN baseline finetuned on NYU v2 dataset

RGB SUPER SLOMO USING HIGH FPS DYNAMIC VISION SENSOR

October 2018 - December 2018 | CMU, Pittsburgh

Developed a deep learning network to produce high frame rate RGB video from a low frame rate RGB video using optical flow derived from high frame rate dynamic vision sensor video as supervision and achieving comparable performance as that of using high FPS RGB video supervision

SMART RECONSTRUCTION

January 2018 - May 2018 | ILIM Lab, CMU, Pittsburgh

Reconstructed traffic from a single stationary camera using keypoint detection and ground plane assumption (similar to the work done by Beyond Pixels paper), tracking and geometric constraints while stabilizing the camera

WEAKLY SUPERVISED OBJECT DETECTION

March 2018 - March 2018 | CMU,

Pittsburgh

Implemented weakly supervised object detection algorithm: WSDDN as part of course project in TensorFlow

SCENE CLASSIFICATION

September 2017 | CMU, Pittsburgh

Implemented scene classification using Spatial Pyramid Matching from scratch as part of course project in Matlab

DIGITAL ART USING SFM

October 2017 - November 2017 | CMU,

Pittsburgh

Developed an application to create portrait effect from single camera using SFM and 3D segmentation as part of course project using OpenCV C++

INTELLIGENT INPAINTING

October 2017 - November 2017 | CMU,

Pittsburgh

Developed an application that removes a person from an image from a single click using pedestrian detection, semantic segmentation and exemplar inpainting in C++

PUBLICATIONS

- Improved ground plane detection in real-time using homography, ICCE [link to the paper](#)
- Understanding performance benefit of asynchronous data transfers in OpenCL programs executing on media processors, HiPC [link to the paper](#)